VUSUMUZI NKOSI

Data Scientist · Forecasting · Fraud & Risk Analytics · Experimentation

Newcastle, KwaZulu-Natal, South Africa · +27 72 641 2519
linkedin.com/in/vusumuzinkosi · github.com/vusumuzinkosi · vusumuzinkosi.github.io

PROFILE

Data scientist and mathematics lecturer with 9+ years of classroom experience, focused on practical, decision-focused modelling, baseline testing and plain-language communication. Recent results: an Eskom demand forecast with 2.4% day-ahead error against Eskom's 2.3%; a fraud review queue for six analysts catching 98% of fraud cases; and a churn call list protecting 82% of value at risk at 10% contact capacity. Currently completing a PGDip in Data Science (STADIO, NQF 8). Seeking data scientist or data analyst roles in South Africa or remote.

TECHNICAL SKILLS

Programming & Data	Python, SQL, R
Machine Learning	scikit-learn, XGBoost, gradient boosting, random forest, logistic regression, K-means, cost-based thresholds, imbalanced classes
Validation	time-based splits, walk-forward testing, leakage checks, baseline comparison
Statistics & Experimentation	A/B testing, power analysis, confidence intervals, hypothesis tests, regression, time-series forecasting, SciPy, statsmodels
Visualisation & Apps	Matplotlib, Seaborn, Streamlit, Excel
Tools	Git, GitHub, Jupyter, Google Colab, VSCode, AWS SageMaker

PROJECTS

Eskom Demand Forecasting — Python · scikit-learn · gradient boosting

- Forecasts South Africa's hourly electricity demand from 4.5 years (39,600 hours) of Eskom data using calendar, holiday and lagged-demand features with 52 weeks of walk-forward validation.
- 3.1% MAPE week ahead — 15% better than baseline — and 2.4% day-ahead error.

Fraud Detection: A Daily Review Queue for Mobile Money — Python · scikit-learn

- Built a review queue from 2.6 million transactions using a time-based split with leakage removed.
- Top 300 daily reviews catch 98% of fraud cases and 89% of money, with 87% genuine flags.

Churn Prediction: Revenue at Risk — Python · scikit-learn · Streamlit

- Business-cost thresholding with a random forest catching 95% of churners (ROC AUC 0.98).
- Expected-loss ranking protects 82% of value at risk at 10% capacity; deployed Streamlit planner.

A/B Test: New Landing Page — Python · statsmodels · SciPy

- Pre-registered two-sided test on 290,584 visitors. Recommended not rolling out: the new page converted 0.16 percentage points lower ($p = 0.19$, CI −0.39 to +0.08).

Customer Segmentation: RFM and K-means — Python · SQL/DuckDB · scikit-learn

- Segmented 5,852 customers using RFM tiers and K-means.
- Loyal big spenders are 19% of customers but 72% of revenue.
- SQL pipeline matched pandas on 776,577 lines; about 1 in 5 new customers reorders within a month.

WORK EXPERIENCE

Special Educator, Mathematics Jan 2012 – Mar 2016
Mdumiseni High School

- Taught mathematics to diverse learner groups, including learners with special needs.
- Developed differentiated lesson plans and assessment rubrics.

Senior Lecturer, Mathematics & IT Feb 2017 – Present
Majuba TVET College

- Analyses learner performance data to identify trends and improve learner outcomes.
- Develops mathematics and IT learning materials and assessments.

EDUCATION

Postgraduate Diploma in Data Science, NQF 8 2026 (in progress)
STADIO Higher Education

- Modules: Introduction to Data Science, Working with Data, Statistical Modelling, Applied Machine Learning, Capstone Project.

Data Science Programme Mar 2025 – Apr 2026
ALX Africa

Postgraduate Diploma in TVET 2024 – 2025
University of the Western Cape

BSc Mathematics & Computer Science 2019 – 2024
University of South Africa

BEd Mathematics & Computer Science 2012 – 2017
University of South Africa

CERTIFICATIONS

- Data Science Certificate, ALX Africa (2025 – 2026)
- Data Analyst Certificate, ALX Africa (2025)
- AI Engineering Bootcamp, Zero To Mastery Academy (2025)
- Prompt Engineering Bootcamp, Zero To Mastery Academy (2025)

KEY STRENGTHS

- Forecasting and time-series modelling with walk-forward validation
- Fraud and risk analytics on large transactional datasets
- Experiment design, power analysis and pre-registered A/B testing
- Translating quantitative results into plain-language business decisions

Vusumuzi Nkosi · Newcastle, KwaZulu-Natal, South Africa · +27 72 641 2519 · linkedin.com/in/vusumuzinkosi · github.com/vusumuzinkosi · vusumuzinkosi.github.io